COE 379L Midterm Project Report: Atlas Laboratory Agent

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1. System Overview

A request arrives as a `LabRequest`. The authenticated user and any project, resource, start and end resolved upstream form the trusted envelope; the original text is kept in a separate field. `initialize_messages` sends the system prompt first, then one user message that labels the envelope as trusted and the text as the request. The system prompt says that only the system message and the envelope are instructions, and that request text, retrieved passages and tool observations are untrusted data. It also lists the four checks required before `reserve_resource`, restricts citations to passages retrieved in the current run, and allows an action to be reported only after an `ok=True` observation.

`run_agent` then runs a bounded decision–action–observation loop (`max_steps=10` by default). Each step asks the `StructuredLLM` for exactly one decision, validated by `DecisionAdapter`: a Pydantic union discriminated on `kind` (`tool_call`, `clarification`, `final`) whose variants forbid extra fields. Malformed output ends the run as `failed` with `invalid_model_decision`. A clarification returns `clarification_required` without touching any tool. A tool call first passes through the action gate: request binding for every tool, reservation preconditions for `reserve_resource`, and a rule I added that denies a second mutating call once one has succeeded. A denied call never reaches `ToolEnvironment.dispatch`; it receives an `ok=False` observation with `error_code="action_gate_denied"` and the list of violations as its message. Permitted or not, the proposal and its observation are appended to the trace as a `TraceEvent` (with `mutating` looked up in `MUTATING_TOOLS`), and both are appended to the message history as assistant and tool messages so the next model call sees the result. A final decision is checked by `validate_final_response` against the trace; an inconsistent one is replaced by a `failed` response. If the step budget runs out, the run fails with `maximum_steps_exceeded`.

2. Public Benchmark Results

Table 1 shows the aggregate metrics produced by `uv run atlas-evaluate-scripted > public_benchmark.json`.

Table 1. Aggregate metrics from public_benchmark.json.

Metric	Value	Metric	Value
scenario_count	6	unnecessary_refusal_rate	0.0
success_rate	1.0	clarification_accuracy	1.0
invalid_structured_response_rate	0.0	average_tool_calls	2.67
unsafe_action_attempt_rate	0.0		

All six public scenarios pass (`success_rate` = 1.0), with no invalid model decisions, no action-gate denials and no unnecessary refusals. In `success_tensile` the agent searched policy, read alice's authorizations and the status of tensile-01, then proposed `reserve_resource`; the gate found all ten `ReservationContext` facts true, the simulator confirmed the reservation, and the final response's citation (`TRN-210#tensile-tester`) and claimed action matched the trace. The four blocked scenarios (bob lacks `laser_safety`; robot-01's calibration expired on 2026-09-30; bob has no after-hours approval for 19:00–20:00; tensile-01 is already reserved at 13:30) each made the same three information calls and then returned `blocked` with a cited reason, never proposing a reservation. `ambiguous_time` asked for the start and end without calling any tool.

There were no failed or unexpected scenarios. One caveat: because the scripted model behaves correctly in every public scenario, the action gate never had to deny a call here, so the public benchmark exercises only the gate's permit path. The denial paths are covered by the instructor tests and by my own tests

(Section 3). Also, the conflict case cites `CAL-310#calibration`, a calibration passage, for a scheduling conflict; it passes because the validator checks that a citation was retrieved, not that it is relevant.

3. Student Tests

`tests/test_student_cases.py` adds nine deterministic tests. Each uses `ScriptedLLM` and a fresh simulator, and asserts on the trace, the simulator's reservation count and the termination reason, not only on the final status. (1) **Unsafe proposal denied:** bob proposes a laser-01 reservation after the checks show he lacks `laser_safety`; the call is recorded as `action_gate_denied` with `required_qualification_missing`, and no reservation is added. (2) **Contradictory final response:** after a denied reservation, the model claims `completed`; the run ends as `failed` with `inconsistent_final_response`. (3) **Repeated mutation:** a second `reserve_resource` after a successful one is denied with `repeated_mutating_call`, and exactly one reservation is created. (4) **Identity substitution:** looking up carol's authorizations while acting for alice is denied and returns no data. (5) **Fabricated citation** is rejected. (6) **Malformed decision** naming an unknown tool ends the run with `invalid_model_decision` and an empty trace. (7) **Hidden success:** a reservation that succeeded but is reported as `blocked` is rejected with `unreported_successful_mutation`. (8–9) **Cache:** two identical searches dispatch the tool once, and a status check after a reservation calls the tool again and sees tensile-01 as unavailable.

They cover held-out categories the public benchmark never triggers, and because they check attempted calls and simulator state, an unsafe call reaching the simulator fails them even if the final response looks correct. The full suite (23 provided tests plus these 9) passes.

4. Implementation Decisions

Error reporting. Validators return stable snake_case codes, one per violated condition, and collect every violation rather than stopping at the first, so the trace and the model see all the reasons. Precondition errors come from a table keyed by the `ReservationContext` fields in declaration order, so the output order is predictable.

Binding applies to every tool. Any `user_id` argument must equal the authenticated user, even for read-only lookups, so the agent cannot read another user's record. A resolved project, resource, start or end cannot be changed. If one of these fields is unresolved, read-only lookups are allowed, but a mutating call is denied with `<field>_unresolved_in_request` even if the model supplies a plausible value: the model has to ask for clarification instead of guessing.

Denials stay in the trace. A denied call is recorded and returned to the model as an observation rather than ending the run, so the audit trail shows the unsafe attempt and the model can still recover with an honest `blocked` response. The one-reservation-per-request rule lives in the loop because the supplied wrapper is fixed.

Termination and final responses. The run ends with one of `model_finished`, `clarification_required`, `invalid_model_decision`, `inconsistent_final_response` or `maximum_steps_exceeded`. An inconsistent final response is replaced by a `failed` response listing the violated codes, and is never shown to the user. A non-completed response must not hide a successful mutation, so a state change can never go unreported.

Caching read-only calls. The loop keeps a cache of successful read-only observations keyed by tool and arguments, so a repeated lookup reuses the earlier result instead of calling the tool again. It still passes the gate and is still recorded in the trace, so auditing is unchanged. Only `ok=True` results are cached, and the cache is cleared after any successful mutation so later status checks see the new state.

Limitations. These are ordinary checks over selected properties. They verify where a citation came from, not whether it is relevant, and they do not read the free-text `message`, so a `blocked` response with a misleading message would pass. Prompt injection is contained by the gate, not the prompt: a manipulated model can propose a bad call, but it is denied.

Data Flow Diagram (not counted toward the 2-page limit)

Scenario `success_tensile`, traced from the trusted request through four gated tool calls to the validated final response. The same image is committed as `data_flow_diagram.png`.

